

Name: _____

Class: _____

Date: _____

Unit 3 Vocabulary

Algorithms & Programming · Study Sheet



Know These Cold

Every term here can show up on the AP exam. Fold the page to hide the definitions and quiz yourself, or quiz a partner.

3.1 Variables and Assignments

Variable

a named storage location for a value that can change

Assignment

storing a value in a variable; replaces whatever was there before

Data type

the kind of value (number, string, Boolean, list); determines what operations make sense

Boolean

a value that is either true or false

String

an ordered sequence of characters

3.2 Data Abstraction

List

an ordered sequence of elements stored under one name

Element

an individual value in a list

Index

a common way to reference elements by their position in a list

Data abstraction

managing complexity by naming a collection of data and hiding the details

Empty list

a list with no elements, written []

3.3 Mathematical Expressions

Algorithm

a finite set of instructions that accomplish a specific task

Sequencing

steps run one after another in order

Selection

choosing which steps to run based on a condition

Iteration

repeating steps

Name: _____

Class: _____

Date: _____

Unit 3 Vocabulary

Algorithms & Programming · Study Sheet

Expression

a value, variable, operator, or procedure call that is evaluated to produce a single value

3.4 Strings

String

an ordered sequence of characters

Concatenation

joining strings together end-to-end

Substring

a part of an existing string

Empty string

a string with no characters, written ""

Length

the number of characters in a string

3.5 Boolean Expressions

Boolean value

a value that is either true or false

Boolean expression

an expression that evaluates to true or false

Relational operator

compares two values ($=$, $\neq$, $>$, $<$, $\geq$, $\leq$) and produces a Boolean

Logical operator

NOT, AND, OR — combines or reverses Boolean values

Equivalent expressions

two expressions that produce the same result for every input

3.6 Conditionals

Selection

determines which parts of an algorithm run based on a condition being true or false

Conditional statement

changes the sequential flow of control by running different statements based on a Boolean expression

Block

a group of statements inside braces { }

Flow of control

the order in which statements are executed

3.7 Nested Conditionals

Nested conditional

a conditional statement placed inside another conditional

Name:

Class:

Date:

Unit 3 Vocabulary

Algorithms & Programming · Study Sheet

Branch

one possible path through a conditional

Compound condition

a condition that combines Booleans with AND / OR / NOT

Decision tree

a diagram that shows each choice as a fork in the path

Equivalent code

different code that produces the same output for all inputs

3.8 Iteration

Iteration

a repeating portion of an algorithm; repeats a set number of times or until a condition is met

Infinite loop

a loop whose ending condition never becomes true

Traversal

accessing each element of a list, one at a time

Counter

a variable that tracks how many times something happens

Accumulator

a variable that builds up a total as the loop runs

3.9 Developing Algorithms

Algorithm

a finite set of instructions that accomplish a task

Pseudocode

a plain-language description of code that isn't tied to one programming language

Flowchart

a diagram of an algorithm using shapes and arrows

Equivalent algorithms

algorithms that give the same output for every input

Building block

an existing, correct algorithm reused to create a new one

3.10 Lists

Traversal

visiting the elements of a list one at a time

Linear search

checks each element in order; works on any list

Name:

Class:

Date:

Unit 3 Vocabulary

Algorithms & Programming · Study Sheet

APPEND

adds a value to the end of a list

INSERT

adds a value at a specific index and shifts the rest right

REMOVE

deletes the element at an index; length decreases by 1

3.11 Binary Search

Binary search

repeatedly checks the middle and eliminates half of a sorted list

Linear search

checks elements one by one in order

Sorted

arranged in order (numeric or alphabetical)

Worst case

the input that requires the most steps

Efficiency

how much time / resources an algorithm uses as input grows

3.12 Calling Procedures

Procedure

a named group of instructions that can be called

Parameter

a placeholder variable in the procedure definition

Argument

the actual value passed in when calling

Return value

the value a procedure sends back to where it was called

Procedure call

a statement that runs a procedure, e.g. `area(3, 7)`

3.13 Developing Procedures

Procedural abstraction

using a procedure by its name and behavior, hiding how it works

Modularity

breaking a program into separate parts / modules

Generalization

using parameters so one procedure works for many inputs

Name:

Class:

Date:

Unit 3 Vocabulary

Algorithms & Programming · Study Sheet

Subproblem

a smaller piece of a larger problem

Duplicate code

copied code that should be replaced by a procedure

3.14 Libraries

Library

a collection of procedures ready to be used in new programs

API

a specification of how a library's procedures behave and are used

Documentation

explains each procedure's name, parameters, and return value

Procedural abstraction

you can use a procedure without knowing how it works

Import

a statement that makes a library's procedures available in your program

3.15 Random Values

Random number generator

produces values that can't be predicted

Inclusive

the range includes both endpoints

Probability

how likely an outcome is, from 0 to 1

Outcome

one possible result of a random event

Pseudorandom

computer-generated numbers that look random but come from an algorithm

3.16 Simulations

Simulation

a program that imitates a real-world process or system

Hypothesis

a prediction that can be tested

Variability

natural randomness or variation in real-world events

Assumption

something the model treats as true to simplify reality

Name: _____

Class: _____

Date: _____

Unit 3 Vocabulary

Algorithms & Programming · Study Sheet

Abstraction

reducing complexity by hiding details and keeping what matters

3.17 Algorithmic Efficiency

Problem

a general description of a task that may or may not be solved with an algorithm

Instance

a problem with a specific input

Decision problem

a problem with a yes/no answer

Optimization problem

a problem that asks for the best solution

Heuristic

an approach that finds a solution that is not guaranteed to be optimal, used when the exact solution is impractical

3.18 Undecidable Problems

Decision problem

a problem with a yes or no answer

Decidable

an algorithm can solve every instance correctly

Undecidable

no algorithm can solve every instance correctly

Halting problem

deciding whether any program will finish running or run forever on a given input

Instance

the problem with one specific input